

PDO Statement	What it Does	What User Gets
<code>new PDO()</code>	Connect	PDO object
<code>prepare()</code>	Prepares query	PDOStatement
<code>bindParam()</code>	Bind variables	Secure input
<code>execute()</code>	Run query	true/false
<code>fetch()</code>	Get 1 row	Array / false
<code>fetchAll()</code>	Get all rows	Array[]
<code>fetchColumn()</code>	Get one value	string/int
<code>lastInsertId()</code>	Last inserted ID	int
<code>rowCount()</code>	Affected rows	int
<code>beginTransaction()</code>	Start transaction	—
<code>commit()</code>	Save	—
<code>rollBack()</code>	Undo	—

✓ PHP + PDO CRUD Example (Full Working Sample)

📌 1. Database Setup (MySQL)

Create a table:

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

✓ 📌 2. db.php — PDO Connection File

```
<?php  
$host = "localhost";  
$db = "test_db";  
$user = "root";  
$pass = "";
```

```
try {
    $pdo = new PDO("mysql:host=$host;dbname=$db;charset=utf8", $user,
$pass);
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
} catch (PDOException $e) {
    die("Connection failed: " . $e->getMessage());
}
?>
```

3. CREATE — Add User

```
<?php
require "db.php";

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $stmt = $pdo->prepare("INSERT INTO users (name, email) VALUES (:name,
:email)");
    $stmt->execute([
        ":name" => $_POST["name"],
        ":email" => $_POST["email"]
    ]);

    echo "User added! ID: " . $pdo->lastInsertId();
}
?>

<form method="POST">
    <input name="name" placeholder="Name">
    <input name="email" placeholder="Email">
    <button type="submit">Add User</button>
</form>
```

4. READ — List All Users

```
<?php
require "db.php";

$stmt = $pdo->query("SELECT * FROM users");
$users = $stmt->fetchAll(PDO::FETCH_ASSOC);
?>

<h2>User List</h2>
<ul>
```

```
<?php foreach ($users as $user): ?>
    <li>
        <?= $user["id"] ?> - <?= $user["name"] ?> (<?= $user["email"] ?>)
        <a href="update.php?id=<?= $user['id'] ?>">Edit</a>
        <a href="delete.php?id=<?= $user['id'] ?>">Delete</a>
    </li>
<?php endforeach; ?>
</ul>
```

5. UPDATE — Edit a User

update.php

```
<?php
require "db.php";

$id = $_GET["id"];

$stmt = $pdo->prepare("SELECT * FROM users WHERE id = :id");
$stmt->execute(["id" => $id]);
$user = $stmt->fetch(PDO::FETCH_ASSOC);

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $stmt = $pdo->prepare("UPDATE users SET name = :name, email = :email
WHERE id = :id");
    $stmt->execute([
        ":name" => $_POST["name"],
        ":email" => $_POST["email"],
        ":id" => $id
    ]);

    echo "Updated!";
}
?>

<form method="POST">
    <input name="name" value="<?= $user['name'] ?>">
    <input name="email" value="<?= $user['email'] ?>">
    <button type="submit">Save</button>
</form>
```

6. DELETE — Remove a User

delete.php

```
<?php
require "db.php";

$id = $_GET["id"];

$stmt = $pdo->prepare("DELETE FROM users WHERE id = :id");
$stmt->execute(["id" => $id]);

echo "User deleted!";
?>
```
